A computer-assisted upper bound of58 for finite-volume hyperbolic Coxeter polytopes

Internal manuscript translation of the frozen proof certificates

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Abstract

We give the final proof composition establishing dimension at most58 for finite-sided, finite-volume real hyperbolic Coxeter polytopes. The argument combines a simultaneous local corner-payment theorem, an exact upper certificate for all finite Coxeter forests of rank59, and a positive shape-curvature surplus at every ideal vertex. Nonsimple cusps are treated through their products of simplex links. The finite upper equals the curvature coefficient, so the strictly positive global correction already gives a contradiction. We specify the computational certificate boundary and distinguish exact arithmetic checking, mathematical review and the ongoing proof-assistant translation. The local geometric atlas and the previously completed dimension59 tail are internal parts of the proof, not additional published assumptions.

1 Statement and scope

Theorem 1. *Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most58.*

No arithmeticity, compactness, simple-cusp or facet-count hypothesis is imposed. The result is an internally checked computer-assisted proof. This manuscript does not assert external refereeing, novelty, optimality, or completed end-to-end Lean verification. The accompanying geometric atlas consists of the frozen local theorem reports and their exact certificates. The endpoint and its recursive dependency manifest are `fresh_a15_d15_dimension58_20261018`.

The proof has two logically distinct dimension steps. The completed internal bound59 excludes all dimensions at least60. We exclude dimension59 here. It would not suffice to exclude59 alone: no unproved claim about induced Coxeter faces is used to supply the higher tail. The later ambient58 search for a bound57 is independent of this proof.

2 Actual corners and cusp multiplicities

We use the norm-two convention for outward roots: diagonal Gram entries are2 and off-diagonal entries are nonpositive. At an actual two-face, the supporting white roots form an elliptic set of size $n-2$. Adjoining either incident side root gives an elliptic edge principal. At a finite vertex, the incident diagram is a finite Coxeter forest of rank$n$.

At an ideal vertex$v$, write its irreducible affine factor ranks as $r_1, \dots, r_s$, with $r_i \geq 1$ and $\sum_i r_i = n-1$. Its Euclidean link is

$$L_v = \prod_{i=1}^{s} \Delta^{r_i}, \qquad V_v = \prod_{i=1}^{s} (r_i + 1).$$

The standard finite-volume link and simple-at-edges facts are the geometric input to the incidence reduction. They do not assert that an ideal vertex itself is simple.

Lemma 2 (Factor-fiber multiplicity)**.** *At an actual ideal two-face corner, its marked pair belongs to the one varying affine factor. A fixed pair in factori occurs in exactly $V_v/(r_i+1)$ actual corner fibers atv. Consequently, if the total pair weight of any rankr affine factor is at most $(r+1)c(r)$, then the actual cusp charge satisfies*

$$C_{\mathrm{act}}(v) \le V_v \sum_i c(r_i).$$

Proof. An edge of the product link varies in exactly one simplex factor. The supporting white diagram retains r_i-1 nodes of that factor and one-node-deleted elliptic pieces from the other factors. The two marked sides restore the whole varying factor. The other factors admit $\prod_{j\neq i}(r_j+1)$ choices, proving the multiplicity.

The use of a whole factor is justified geometrically. A positive kernel combination of an affine subdiagram is a nonzero feasible null vector: it is nonzero because its product with a strict chamber center is negative, and is feasible by acuteness. Its projective point is an ideal vertex, by strict convexity of the closed Klein ball. Within the ideal diagram it is a complete irreducible affine component, because a proper principal subdiagram of such a component is positive definite. The internal factor-fiber report records this argument and the corresponding published parabolic-containment input. □

3 The deficit and its shape-curvature lower bound

For a two-face$F$, let$s(F)$ be its number of sides and put

$$D_5(P) = \sum_F (5 - s(F)).$$

For a vertex figure of dimension$d = n-1$, define

$$H(t) = \sum_{j=0}^{d} h_j t^j, \quad V = H(1), \quad G = \sum_{j=0}^{d} h_j \min(j, d-j), \quad \kappa = \frac{5G - dV}{2}.$$

For an ideal vertex, $H(t) = \prod_i (1 + t + \cdots + t^{r_i})$. For a finite vertex, the figure is a simplex, so $h_j = 1$.

Lemma 3 (Shape-curvature inequality)**.** *In dimension59,*

$$D_5(P) \ge \frac{783}{2} v_f + \sum_{v \ ideal} \kappa(L_v) + 4350.$$

Proof. Choose a generic linear function on the bounded projective closure. For each vertex figure let$E$ be its number of edges andE^0 the number not cut by the induced level section. Khovanskii's exact reduction gives the two-face incidence sum $I = \sum E$ and $2f_2 = \sum E^0$. Hence

$$D_5 = \sum_v \left(\frac{5}{2} E_v^0 - E_v \right).$$

Orient each simple vertex figure generically. A vertex of index$j$ has j descending and$d-j$ ascending edges, and the index distribution is h_j. Assign an uncut edge below the section to its upper endpoint and one above to its lower endpoint. Each vertex receives either$j$ or$d-j$ such contributions, giving $E^0 \geq G$.

The global minimum and maximum of the original polytope are distinct; their level sections miss the entire vertex figures. At either one, symmetry and the index formula give

$$E - G = \sum_{j>d/2} h_j(2j-d).$$

All the vertex figures in question are simplices or products of simplices, so every$h_j \geq 1$. At$d = 58$ each extremum contributes an extra at least $\sum_{j=30}^{58}(2j-58) = 870$ to$E^0 - G$. Multiplication by$5/2$ and addition of the two extrema gives4350, counted once.

For a finite vertex $G = \sum_{j=0}^{58} \min(j, 58-j) = 841$ and $E = 58 \cdot 59/2$. Thus its curvature is

$$\frac{5 \cdot 841 - 58 \cdot 59}{2} = \frac{783}{2}.$$

The ideal term is the statedκ. This proves the inequality. □

The exact incidence and index identities are published inputs. The retained shape term and the quantitative two-extrema strengthening above are internal deductions, not assumptions of the dimension bound.

4 One simultaneous local payment rule

A corner payment is an intrinsic nonnegative function of the actual marked finite diagram or varying affine factor, including any stated whole-component presence conditions. Its definition is the frozen baseline in `fresh_a14_cd16_ambient59_frontier_20261018`, followed by the changes below. That baseline has its own complete local proof atlas, with selected and nonselected routes, arbitrary supporting-root arguments and a fixed actual-corner priority order.

Proposition 4 (Final local theorem)**.** *The final simultaneous rule gives every triangular two-face payment at least*2 *and every quadrilateral payment at least*1*. Relative to the baseline, whole A15 pairs*$(0,12),(2,14)$ *and whole D15 pair*$(0,11)$ *change from*$1/2$ *to*0*. Add the following whole-component increments:*

Small marked role	*Large marked role*	*Increment per pair*
A1	*A12 endpoint*	$1/4$
A1	*A13 second node*	$1/4$
A4 inner node	*A13 second node*	$1/2$
A4 endpoint	*A11 endpoint*	$1/2$

All ideal weights and all other finite weights remain unchanged.

Proof. Apply the independently proved ideal and finite6 actual-corner priorities first. Their short5, protected-leaf, whole-small and typed outputs survive. The cycle11 exception uses its separate replacement theorem; a short ideal path is not silently presumed to be a unit. The independent short8-to-short7-to-short6 chain likewise terminates in retained outputs.

In the remaining branch, the complete finite selected proof either chooses an unchanged contribution or a changed rank15 pair. The full eligible domain has536 patterns. Explicit induced native embeddings give482 A15 maps and870 D15 maps, including the failed-search patterns. The eight changed A15 maps lie in four contexts, and the ten changed D15 maps in five contexts. The finite lists are complete because of the inherited universal reduction, not because a selection of known faces happened to contain no other example.

For A15, two Lorentz cores exclude all remaining contacts by projection squares at least4. Two radical cores force either complete orthogonality or a retained short5 path. Every paying factor is therefore whole. Each context retains a distinct A16 half at$(1, 13)$ or$(2, 14)$. Two contexts add two quarters at two different other corners; the other two use the A4/A13 half. Thus the lower total remains1.

For D15, proper contexts1030 and1613 require a norm-two root whose projection square is at least6. In context1641 the forced new center is saturated: its residual vector is zero. Its product with a required simple fork leaf is positive, zero or at most-2, never-1; the completion is impossible. In whole contexts1682 and2501 the relevant inverse diagonal entries4,4,14 seal the core unless the independent short8 alternative applies. A retained D16$(1, 12)$ half and a distinct new A4/A11 endpoint half give total1.

The geometric reasoning used here has no spectator cutoff. A Gram radical of an indefinite Lorentz core represents the zero vector, because the orthogonal complement of a timelike span is positive definite. Nonpositive contact products must annihilate that radical. The checked sign domains force a retained short path or orthogonality to the whole core. For a nondegenerate core, the residual square is $2 - q^t G^{-1} q \geq 0$; equality forces the root into the core span. These arguments prove whole-factor and proper-completion statements before using the displayed finite alternatives.

No new zero is used as another new proof's positive output. The A15$(1, 13)$ unit needed by the A14 theorem, the D15$(1, 12)$ half needed by the D14 theorem and the D15$(0, 10)$ unit needed by the ideal15 theorem remain. The A1/D15 node11 sixteenth is a separate cross-component role. All rank16/17 and contextual alternatives are retained. The complete low-core, disjoint, protected and adaptive routes have no new long-rank15 output. The216 exact literal checks supplement these universal arguments and have zero new debit or credit; they are not a universal face census. Elementary triangle contributions are disjoint from the new long zeros and retain minimum2. □

The complete proofs are the A15 three-coupon and D15 A4/A11 reports. Their supporting-root and route-completeness arguments are internal theorems. In a formalization they must be translated, not admitted as external axioms under the label "computer verification."

5 Exact finite-forest upper bound

Proposition 5. *Every actual finite Coxeter forest of total rank59 has final charge at most*783/2.

Proof. The final ordinary costs are A15=D15=100 and B15=97. The four added component-pair costs are respectively$1/2, 1/2, 2, 2$. The representative A1/D13 allowance becomes33/16, and A4/D13 becomes8. The A1/A12 and A4/A11 charges fit their unchanged allowances5/2 and8. The exact certificate checks all180 actual component types through rank59, their bases and all32,400 ordered pair comparisons simultaneously. The arbitrary-label rank-two family and exceptional finite types are included by the stated representative classification.

Fix all multiplicities of ranks9 through16 and of the separate A17 representative. There are961 possible large-component vectors. Fixing the rank5 multiplicity as well gives2574 expanded vectors and interaction keys. Residual ranks are1–4,6–8,17–59; residual rank17 is D17, whereas A17 has

already been fixed. Residual pairwise upper interactions vanish. Each residual item consists of its base plus all interactions with the fixed factors;128,700 exact identities check this statement for the entire domain.

For positive-rank items(r, w_r), a table$V(0) = 0$ satisfying

$$V(k) \geq V(k-r) + w_r \qquad (r \leq k)$$

dominates every additive residual forest by induction after removing one item. An attaining predecessor proves equality when required. The5148 normal/no-A2 tables satisfy6,723,288 such inequalities and 303,732 attaining-predecessor checks. The verifier checks inequalities and exact equality rather than trusting an optimizer's status.

There are two contextual charges. The old A2/A2 charge is $a_2(a_2-1)/4$ when a whole A10 and at least two A1 factors occur. The other charge is$a_2 d_5$ when a whole A9 and at least one A1 occur. The representative guards may allow any rank10/rank9 representative and replace D5 by A5. All A2 multiplicities are fixed explicitly. The no-context, one-reserved-A1 and two-reserved-A1 branches, using no-A2 residual tables, cover every forest. The last branch includes both charges if both guards hold. The relaxed one-A1 branch permits extra A1 only as an upper relaxation; the reconstructed winning forest has its full contextual cost checked. There are9191 options in total.

Every expanded upper is at most6264 sixteenths, namely783/2. This is a representative upper, not a claim of actual attainment. Its representative maximizer $8A_1 + 3A_5 + 3D_{12}$ has actual literal cost355.5. Actual sharpness is unnecessary for the conclusion. □

6 The ideal envelope and complete terminal domain

The normalized affine cost$c(r)$ is the total factor-pair allowance divided by$r+1$. For the completed58 proof it is

r	10	11	12	13	14	15	16
$c(r)$	53/11	31/6	11/2	40/7	367/60	411/64	227/34

with$c(r) = r/2$ for$1 \leq r \leq 9$. For$r \geq 17$ the exact all-family envelope through58 is in the exported certificate; it follows the proved classical graph-count formulas, not a sampled large-rank cutoff. The factor-fiber lemma gives $C_{\mathrm{act}} \leq C_{\mathrm{up}} = V \sum c(r_i)$.

Lemma 6 (Low-nine concentration)**.** *Every positive rank partition of58 can be reduced to one with at most one part below9, such that its normalized curvature-minus-envelope margin and its product$V$ do not increase.*

Proof. For$1 \leq a \leq b < 9$, replace(a, b) by$(a-1, b+1)$, omitting zero. LetU_r be uniform on$\{0, \ldots, r\}$. Before centering, the probability of$U_a + U_b = j$ is

$$\frac{\min(j+1, a+1, b+1, a+b-j+1)}{(a+1)(b+1)}.$$

After the replacement the denominator is$a(b+2)$, smaller by$b-a+1$. The new probability is larger on the tails$j < a$ or$j > b$ and smaller on$a \leq j \leq b$. Both distributions are symmetric about the same mean. Symmetrizing a convex test function makes it nondecreasing with distance from that mean; subtracting its value at the crossing radius proves the convex-order inequality. Adding the other independent uniforms preserves it.

For$d = 58$, the normalized curvature is

$$\frac{\kappa}{V} = \frac{3d}{4} - \frac{5}{2}\mathbb{E}\left|\sum_i U_{r_i} - \frac{d}{2}\right|.$$

It therefore does not increase. The normalized cost is unchanged, because both affected ranks stay at most9 and$c(r) = r/2$ there. The product factor decreases from$(a+1)(b+1)$ to$a(b+2)$. Each move increases the bounded integer$\sum r_i^2$, so iteration terminates. No concentration of the ranks at least9 is used. □

Proposition 7. *Every actual ideal vertex in dimension59 has* $\kappa - C_{\mathrm{act}} \geq 91/2$.

Proof. Enumerate every multiset of ranks at least9 summing to$58-\ell$, for $0 \leq \ell \leq 8$, adjoiningℓ if it is positive. This gives exactly 1890 terminal profiles. For every profile the exact certificate checks the product polynomial, its integer coefficients and moments

$$V = \sum_j h_j = \prod_i (r_i + 1), \qquad G = \sum_j \min(j, 58-j) h_j,$$

and the rational normalized margin

$$m = \frac{5G - 58V}{2V} - \sum_i c(r_i).$$

All are positive. The minimum normalized margin is$1/340$, attained at$(9,9,9,15,16)$; the minimum absolute surplusVm is$91/2$, attained at the singleton profile(58).

Under Lemma 6, the original profile has $V \geq V'$ and$m \geq m' > 0$. Hence$Vm \geq V'm' \geq 91/2$. This multiplication uses positivity; it would not follow from negative terminal margins. Finally $C_{\mathrm{act}} \leq C_{\mathrm{up}}$ gives the same lower surplus for the actual cusp charge, with its correct factor multiplicities. Nonsimple cusps have not been discarded. □

7 The contradiction

For any polygon with at least five sides the deficit$5-s$ is nonpositive. Proposition 4 therefore gives, by summing nonnegative corner weights and then regrouping by vertices,

$$D_5 \leq \frac{783}{2} v_f + \sum_{v \text{ ideal}} C_{\mathrm{act}}(v).$$

Together with Lemma 3 and Proposition 7, this implies

$$0 \geq 4350 + \sum_{v \text{ ideal}} \big(\kappa(L_v) - C_{\mathrm{act}}(v)\big) \geq 4350 + \frac{91}{2} v_i > 0.$$

Thus dimension59 is impossible, including the compact case$v_i = 0$. The completed internal dimension59 theorem excludes all dimensions at least60. This proves Theorem 1.

8 Certificates, assumptions and formalization

The exact endpoint is reproduced by

```
python3 scripts/fresh_a15_d15_dimension58_20261018.py --verify
```

It verifies the common frozen local/count/baseline bindings, complete saved domains, polynomial moment identities and curvature arithmetic. It does not repeat the historical native census, all supporting-root classifications, convolution generation or Bellman optimization. The finite-count producer's separate replay checks its full Bellman certificates. The geometric producers reconstruct their physical cores and exact proper-completion or sealing calculations; their reports contain the universal arguments not expressible as finite absence tests.

The plain export `data/certificates/finite_volume58_arithmetic_20261018.json` contains integer sixteenth-unit tables, typed cost matrices, complete fixed/context rows,1890 polynomial coefficient lists and rational envelopes, with source hashes. It is explicitly an export, not itself a formal proof. A kernel checker must establish the finite-domain bijections, domination, recurrence inequalities, contextual coverage, polynomial products and exact margins. Hash agreement establishes only identity of inputs.

The published assumption boundary consists of precisely stated finite/affine Coxeter classification facts, the Lannér and quasi-Lannér order ceilings, the standard finite-volume cusp structure and the incidence/index identities used above. The high-rank internal census reproduces26 classical quasi-Lannér diagrams from146,262 attachments; the published order-ten ceiling is a separate input. All new local transfer theorems, arbitrary-support seals, priority retention, complete typed budgets, factor-fiber bookkeeping, concentration and completed dimension conclusions are internal obligations. They are not to be axiomatized wholesale in a proof assistant.

The companion map `docs/finite_volume58_lemma_map.md` supplies quantified interfaces for these obligations. The parent Lean project currently reports kernel-checked generic additive-table induction, selected-deficit summation, integer-gap and tail-composition results. The complete geometric-to-arithmetic bridge and all endpoint data have not yet been connected into an end-to-end Lean proof of58. Accordingly the formalization remains partial, while the frozen mathematical proof has completed generation/replay and three full internal composition reviews. Those reviews did not independently reprove every historical classification or repeat every numerical certificate.

A Minimal frozen proof-atlas index

Role	Artifact stem (each has source, JSON and report)
Final composition	`fresh_a15_d15_dimension58_20261018`
Finite upper and ideal rows	`fresh_a15_d15_ambient59_forest_closure_20261018`
Simultaneous baseline	`fresh_a14_cd16_ambient59_frontier_20261018`
A15 final transfer	`fresh_finite_a15_three_coupon_zero_transfer_20261018`
D15 final transfer	`fresh_finite_d15_a4_a11_half_zero_transfer_20261018`
Higher tail	`fresh_depth6_dimension59_20261018`
Factor geometry	`fresh_cusp_factor_payment_framework_20261017`
Shape curvature	`fresh_cusp_link_shape_curvature_20261017`
Convex order	`fresh_distance14_unrestricted_dimension156_20261017`
Classical high-rank census	`fresh_high_rank_quasilanner_census_20261017`

The recursive source manifests, rather than this short index, identify all inherited dependencies. The companion reproduction guide gives commands and verification limits. Later ambient58 experiments are not part of this index and cannot silently replace these frozen inputs.

References

[1] A. G. Khovanskii, *Hyperplane sections of polyhedra, toroidal manifolds, and discrete groups in Lobachevskii space* (1986), Theorems2,6,9, as used in the frozen shape-curvature report. Primary text: `https://www.math.utoronto.ca/askold/1986-Faa-1-english.pdf`.

[2] M. Chein, *Recherche des graphes des matrices de Coxeter hyperboliques d'ordre*$\leq$ 10 (1969); order ceiling and classification tables cited in the frozen high-rank census. Primary text: `https://www.numdam.org/item/M2AN_1969__3_3_3_0.pdf`.

[3] A. Felikson and P. Tumarkin, *On simple ideal hyperbolic Coxeter polytopes*, arXiv:math/0502413; the parabolic-subdiagram containment statement in Section1 is cited by the factor-fiber report. Primary text: `https://arxiv.org/pdf/math/0502413`.